Tree-exact diffusion scores for non-Gaussian Markov chains: what second-order message closure computes

Giovanni Mantovani
Department of Computing Sciences
Bocconi University
giovanni.mantovani@studbocconi.it

Abstract

Over an unrestricted function class, the global minimiser of the objective that trains a diffusion model is the score of the *empirical* distribution, not of the data distribution — a model attaining it exactly, integrated to zero noise, would reproduce its training set and nothing else. Deciding what a network actually learned means comparing it against the score it should have learned, which for realistic data is unavailable. We exhibit a family where it is available and non-trivial: a first-order Markov chain with continuous, non-Gaussian transitions. Coordinate-wise noising multiplies the prior by *unary* factors, so the posterior factor graph is still the prior's chain — a tree, on which sum-product is exact. The score is therefore available as an exact functional recursion for a correlated, non-Gaussian model, and numerically to a validated tolerance once the continuous messages are discretised — we validate that grid representation against the Gaussian closed form and against brute-force enumeration of the discretised posterior. We then prove that single-Gaussian message closure computes *exactly* the LMMSE estimator of the covariance-matched Gaussian model, so the gap between it and full sum-product is precisely the price of discarding everything beyond second moments — not an uncontrolled approximation but a named estimator. The same posterior statistics also supply the exact likelihood gradient needed to learn the transition, which the longer paper develops.

1 The problem

Diffusion models are trained by regressing the posterior mean of a clean signal given its noised version [1–4]. A network never sees the data distribution P_0; it sees M samples, and the global minimiser of that regression is the score of the empirical measure $\hat{P}_0 = M^{-1} \sum_\mu \delta(\cdot - \mathbf{a}^\mu)$ smoothed by the forward Gaussian. A model attaining it exactly would generate its training set and nothing else. That diffusion models instead generate new data is the memorisation–generalisation question studied in the statistical mechanics of these models [5, 6].

Progress on it needs a target one can compute. For realistic data the exact score is unavailable, so a network's error cannot be separated from the difficulty of what it was approximating. Tractable families exist but the most common are Gaussian, where the score is linear and every question about structure has a degenerate answer. Gaussian mixtures keep a closed-form score, but it is nonlinear only in the mixture index, not in any dependence between coordinates.

This work. We use a first-order Markov chain: diffusion is applied to sequential data, and studying how structure is exploited requires data that has structure one can dial. The chain is variance-matched, so Gaussian, Laplace, Student, uniform and bimodal innovations share a covariance exactly and differ only beyond second moments. Two things follow, and this note is about the first: the score is computable by local inference — exactly as a functional recursion, and to a measured tolerance once discretised — and the standard second-order approximation to that inference turns out to be an identifiable estimator rather than an uncontrolled one. The same machinery also learns the kernel when it is unknown, which the longer paper develops.

2 The model, and why its score is computable

Forward process. With η standardised Gaussian white noise, $\langle \eta(t)\eta(t')\rangle = \delta(t-t')$, the forward Ornstein–Uhlenbeck process and its solution are

$$\frac{\mathrm{d}\mathbf{x}}{\mathrm{d}t} \;=\; -\mathbf{x} + \sqrt{2}\,\eta(t), \qquad \mathbf{x}(t) \;=\; e^{-t}\mathbf{a} + \sqrt{\Delta_t}\,\mathbf{z}, \qquad \Delta_t = 1 - e^{-2t}, \quad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})\,. \tag{1}$$

The reverse-time convention is fixed once in Appendix A and used without variation.

Prior. P_0 factorises as

$$P_0(\mathbf{a}) \;=\; \mu(a_1)\prod_{i=2}^{L} K(a_i \mid a_{i-1}), \qquad K(a' \mid a) \;=\; \varphi(a' - \rho a), \tag{2}$$

with φ an innovation density of mean zero and variance q, and $|\rho| < 1$. We draw $a_1 \sim \mathcal{N}(0,1)$ and set $q = 1 - \rho^2$, which makes $\mathrm{Var}(a_i) = 1$ at every site and $\mathrm{Cov}(a_i, a_j) = \rho^{|i-j|}$ exactly, *whatever the shape of* φ. Variance matching is what makes the family a controlled probe: Gaussian, Laplace, Student, uniform and bimodal innovations share the same covariance and differ only beyond second moments. The chain is covariance-stationary but not strictly stationary; Appendix C states the difference, and shows why it collapses in the Gaussian case.

The score is a posterior mean. From (1), P_t is a Gaussian convolution of P_0. Differentiating under the integral and dividing by $P_t(\mathbf{x})$ — the Gaussian factor over $P_t(\mathbf{x})$ is exactly the posterior density $P(\mathbf{a} \mid \mathbf{x}, t)$ — gives

$$S(\mathbf{x}, t) \;=\; -\frac{\mathbf{x} - e^{-t}\,\langle \mathbf{a}\rangle_{x,t}}{\Delta_t}, \qquad \langle \mathbf{a}\rangle_{x,t} = \int \mathrm{d}\mathbf{a}\,\mathbf{a}\,P(\mathbf{a} \mid \mathbf{x}, t). \tag{3}$$

Equation (3) is a change of variables, not an approximation: computing the score and computing the posterior mean are the same problem. The identity is classical in the empirical-Bayes literature [7–9] and underlies denoising score matching [4, 10]. For squared loss $\langle \mathbf{a}\rangle_{x,t}$ minimises the conditional risk (Appendix B), so it is the reference every estimator below is scored against.

The likelihood factors are *unary*: the forward noise is independent across coordinates, so each observation touches one latent variable. Unary factors reweight node potentials without creating edges among the a_i, so conditioning leaves the factor graph a chain — a tree, on which sum-product returns exact marginals [11–13]. On a chain these are the forward–backward recursions [14, 15], reducing to the Kalman filter when everything is Gaussian [16, 17].

Exactness is at the level of *functional* messages. A message is a function on $\mathbb{R}$, forced by the state space, so an implementation must represent one; we use a truncated grid of N_g points, which makes the recursion exact for a finite-state model whose discretisation error is measured separately. At $N_\mathrm{g} = 401$ it reproduces the Gaussian closed form to 9.2×10^{-15} and brute-force enumeration of the discretised posterior to 1.6×10^{-14}. Cost is $O(LN_\mathrm{g}^2)$ per sequence against $O(N_\mathrm{g}^{L-1})$ for direct summation.

What the standard Gaussian approximation computes. Propagating only the first two moments of each message is the usual approximation. On this model it has an exact characterisation, which turns a heuristic into a measurement:

Proposition 1 (Gaussian closure computes the LMMSE estimator)**.** *Let $\Pi_\mathrm{G} f$ be the Gaussian density matching the mean and variance of $f/\int f$, and let* moment-projected sum-product *be the forward–backward recursion applying Π_G to each message immediately after every transition integral, multiplying by the site's unary likelihood before the next. Take the chain* (2) *with*

$a_i = \rho a_{i-1} + \varepsilon_i$, $\rho \neq 0$, *Gaussian unary likelihoods,* $\mathbb{E}[a_1] = 0$, *an innovation law* φ *with mean zero and finite variance* q, *and an initial message Gaussian or projected by* Π_{G}. *Then it returns the LMMSE estimator of the covariance-matched Gaussian model,*

$$\widehat{\mathbf{a}}_{\mathrm{LMMSE}} = \mathrm{Cov}(\mathbf{a}, \mathbf{x})\ \mathrm{Cov}(\mathbf{x})^{-1}\mathbf{x} = e^{-t}\mathbf{\Sigma}_0 \left(e^{-2t}\mathbf{\Sigma}_0 + \Delta_t \mathbf{I}\right)^{-1} \mathbf{x}, \tag{4}$$

and returns the same estimator for every innovation law sharing the moments $(0, q)$.

So the gap between moment-projected and full sum-product is precisely the price of replacing the prior by its covariance twin — not an artefact of a message representation. It vanishes as t grows, because the noised marginal P_t Gaussianises and Tweedie's prefactor α_t/Δ_t attenuates the score's sensitivity to structure beyond second moments — not because the latent posterior becomes Gaussian, which it does not: as $t \to \infty$ the observation becomes uninformative and $P(\mathbf{a} \mid \mathbf{x}, t)$ approaches the (non-Gaussian) prior. On the Laplace chain the median relative score error falls from 0.241 at $t = 0.05$ to 7.3×10^{-6} at $t = 3$ (Figure 1). The hypotheses and the proof are in Appendix E.

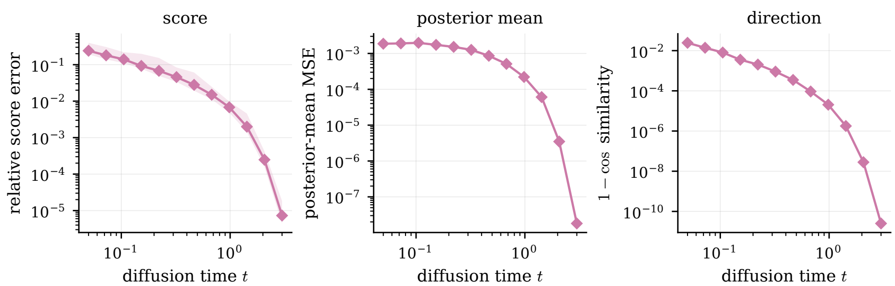

Figure 1: What the Gaussian second-order baseline costs, against sum-product under the true kernel. By Proposition 1 this gap is exactly the price of replacing the prior by its covariance-matched Gaussian, so it measures non-Gaussianity rather than approximation quality. Laplace chain, $\rho = 0.85$; median over seeds with the 10–90 percentile band.

3 The same inference also learns the chain

Everything above assumes the kernel is known. It need not be. The pairwise posterior marginals that sum-product already produces are exactly the expected transition statistics that likelihood learning needs: writing $\mathcal{L}(\theta) = \sum_\mu \log P_{t,\theta}(\mathbf{x}^\mu)$, Fisher's identity gives $\nabla_\theta \mathcal{L}$ as a posterior expectation of the complete-data score, so one forward–backward pass yields the exact gradient and no differentiation through the $2(L-1)$ message updates is required. Accumulated over edges the pairwise beliefs form a single matrix $\mathbf{\Xi}$ that is a complete summary for the resulting surrogate objective, so the maximisation step never revisits the data; for Gaussian innovations it is closed form, the belief-weighted Yule–Walker equations. The derivations, the mixture-innovation extension, and the recovery experiments are in the longer paper.

4 Scope, and what is open

Exactness is structural, not numerical: the recursion is exact on the functional messages, and what an implementation actually runs is a truncated grid whose truncation and quadrature error must be measured rather than assumed. The figures quoted above are agreements at the stated configuration, not a bound for arbitrary kernels, tails or noise levels — trapezoidal quadrature is second order, and it degrades when a likelihood is narrower than a few grid cells. Everything here also assumes exact first-order Markov structure; under a non-Markov prior the inference is misspecified in a way more data does not repair. Inference costs $O(LN_{\mathrm{g}}^2)$ per sequence, well above a network forward pass, so what this model buys is a computable target, not speed.

The open question is architectural. Sum-product on a chain suggests a network narrow but as deep as twice the chain, unlike the tree-structured case where the natural depth is the tree's [18, 19]. Which architectures exploit Markovianity, and at what depth, is what this setting is built to measure.

References

[1] Jascha Sohl-Dickstein, Eric Weiss, Niru Maheswaranathan, and Surya Ganguli. Deep unsupervised learning using nonequilibrium thermodynamics. In *Proceedings of the 32nd International Conference on Machine Learning (ICML)*, pages 2256–2265, 2015.

[2] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 33, pages 6840–6851, 2020.

[3] Yang Song, Jascha Sohl-Dickstein, Diederik P. Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations. In *International Conference on Learning Representations (ICLR)*, 2021.

[4] Pascal Vincent. A connection between score matching and denoising autoencoders. *Neural Computation*, 23(7):1661–1674, 2011.

[5] Giulio Biroli and Marc Mézard. Generative diffusion in very large dimensions. *Journal of Statistical Mechanics: Theory and Experiment*, 2023(9):093402, 2023.

[6] Giulio Biroli, Tony Bonnaire, Valentin de Bortoli, and Marc Mézard. Dynamical regimes of diffusion models. *Nature Communications*, 15:9957, 2024. doi: 10.1038/s41467-024-54281-3.

[7] Herbert Robbins. An empirical Bayes approach to statistics. In *Proceedings of the Third Berkeley Symposium on Mathematical Statistics and Probability*, volume 1, pages 157–163, 1956.

[8] Koichi Miyasawa. An empirical Bayes estimator of the mean of a normal population. *Bulletin of the International Statistical Institute*, 38(4):181–188, 1961.

[9] Bradley Efron. Tweedie's formula and selection bias. *Journal of the American Statistical Association*, 106(496):1602–1614, 2011.

[10] Aapo Hyvärinen. Estimation of non-normalized statistical models by score matching. *Journal of Machine Learning Research*, 6:695–709, 2005.

[11] Judea Pearl. *Probabilistic Reasoning in Intelligent Systems*. Morgan Kaufmann, 1988.

[12] Frank R. Kschischang, Brendan J. Frey, and Hans-Andrea Loeliger. Factor graphs and the sum–product algorithm. *IEEE Transactions on Information Theory*, 47(2):498–519, 2001.

[13] Marc Mézard and Andrea Montanari. *Information, Physics, and Computation*. Oxford University Press, Oxford, 2009.

[14] Leonard E. Baum, Ted Petrie, George Soules, and Norman Weiss. A maximization technique occurring in the statistical analysis of probabilistic functions of Markov chains. *The Annals of Mathematical Statistics*, 41(1):164–171, 1970.

[15] Lawrence R. Rabiner. A tutorial on hidden Markov models and selected applications in speech recognition. *Proceedings of the IEEE*, 77(2):257–286, 1989.

[16] Rudolph E. Kalman. A new approach to linear filtering and prediction problems. *Journal of Basic Engineering*, 82(1):35–45, 1960.

[17] Herbert E. Rauch, Frank Tung, and Charlotte T. Striebel. Maximum likelihood estimates of linear dynamic systems. *AIAA Journal*, 3(8):1445–1450, 1965.

[18] Song Mei. U-nets as belief propagation: Efficient classification, denoising, and diffusion in generative hierarchical models, 2024. arXiv:2404.18444.

[19] Jérôme Garnier-Brun, Marc Mézard, Emanuele Moscato, and Luca Saglietti. How transformers learn structured data: insights from hierarchical filtering, 2024.

[20] Brian D. O. Anderson. Reverse-time diffusion equation models. *Stochastic Processes and their Applications*, 12(3):313–326, 1982.

[21] Ulrich G. Haussmann and Etienne Pardoux. Time reversal of diffusions. *The Annals of Probability*, 14(4):1188–1205, 1986.

A Notation

L is the chain length, which is also the ambient dimension; M is the number of observed sequences; N_{g} is the number of quadrature points; K is always the Markov transition kernel, φ the innovation density, ρ the autoregressive coefficient and q its variance. Vectors are bold lowercase, matrices bold uppercase. The forward process is $\mathbf{x}(t) = e^{-t}\mathbf{a} + \sqrt{\Delta_t}\,\mathbf{z}$ with $\Delta_t = 1 - e^{-2t}$; the reverse SDE and its sign conventions follow Anderson [20], Haussmann and Pardoux [21] and are stated in full in the companion paper.

B The score–posterior identity

P_t is a Gaussian convolution of P_0, and $\nabla_{\mathbf{x}} g_{\Delta_t}(\mathbf{x} - e^{-t}\mathbf{a}) = -\frac{\mathbf{x}-e^{-t}\mathbf{a}}{\Delta_t} g_{\Delta_t}(\mathbf{x} - e^{-t}\mathbf{a})$, with the integrand dominated uniformly whenever P_0 has a finite first moment. Dominated convergence then licenses differentiation under the integral; dividing by $P_t(\mathbf{x})$ and recognising the posterior density gives (3). This is Tweedie's formula [7–9]. Since $\langle \mathbf{a} \rangle_{x,t}$ minimises the conditional squared risk, every error below is reported *against* it rather than as raw risk: the Bayes floor is common to all estimators, and subtracting it is what makes the comparison informative.

C Stationarity

With $q = 1 - \rho^2$ and $\mathrm{Var}(a_1) = 1$, second moments propagate exactly: $\mathrm{Var}(a_i) = 1$ and $\mathrm{Cov}(a_i, a_j) = \rho^{|i-j|}$ for every innovation shape. The chain is therefore *covariance*-stationary, which is what makes the variance-matched family a controlled probe. It is not strictly stationary for non-Gaussian φ — the invariant law of $a_i = \rho a_{i-1} + \varepsilon_i$ is that of $\sum_{k\geq 0} \rho^k \varepsilon_{-k}$, which is not Gaussian — and we do not claim it is. The two notions coincide for Gaussians because a Gaussian law is determined by its first two moments.

D Protocol

The reported numbers come from one frozen configuration, defined in a single file and imported by the experiments that produce them: Laplace chain unless stated, $\rho = 0.85$, $L = 32$, $N_{\mathrm{g}} = 401$, $A = 8$, twelve log-spaced noise levels on $[0.05, 3.0]$, budgets $M \in \{32, \ldots, 4096\}$, and 256 held-out sequences on a disjoint seed. The ring uses $T = 12$. Both arms select their optimisation budget on a disjoint validation bundle; EM's own internal stop is a signed log-evidence tolerance, never ρ, since a settled ρ certifies a converged-looking trace over a kernel that is nowhere near converged. The configuration file also declares a shape-based threshold; nothing reads it, and no reported number was produced by it (see the longer paper's protocol appendix).

E Proposition 1: hypotheses and proof

Remark 1 (On the hypotheses)**.** *Four are load-bearing and worth separating.* Finite second moments *are what make* Π_{G} *defined at all.* Zero mean *is not cosmetic: the general LMMSE carries* $\mathbb{E}[\mathbf{a}] + \mathrm{Cov}(\mathbf{a}, \mathbf{x})\,\mathrm{Cov}(\mathbf{x})^{-1}(\mathbf{x} - \mathbb{E}[\mathbf{x}])$*, and* (4) *is the centred special case.* $\rho \neq 0$ *is required because the backward map of the Gaussian closure recursion divides by* ρ*; at* $\rho = 0$ *the chain has no edges, the sites are independent, and the claim holds trivially and separately, each site's posterior mean being the scalar Gaussian* $e^{-t}\mathbf{\Sigma}_0(e^{-2t}\mathbf{\Sigma}_0 + \Delta_t)^{-1}x_i = e^{t}\mathbf{\Sigma}_0(\mathbf{\Sigma}_0 + \Delta_t e^{2t})^{-1}x_i$. Where the projection is applied *matters because* Π_{G} *does not commute with multiplication by the likelihood; Proposition 1 fixes the order, and a different order is a different algorithm.*

The statement is about exact message functions on $\mathbb{R}$ *and an analytic projection operator.* ***The grid plays no part in it.*** *Discretisation enters only when the* unrestricted *non-Gaussian messages are represented numerically, which is what the recursion is compared against; conflating the two would turn a theorem about a closure into a claim about quadrature.*

Proof. Write $\mu_i^{\rightarrow}$ for the forward message at site i after projection and ℓ_i for the Gaussian unary likelihood. We show by induction that every projected message coincides with the corresponding exact message of the covariance-matched Gaussian chain — the chain with the same ρ and innovation variance q but Gaussian innovations.

Base. The initial message is Gaussian by hypothesis, or is projected to the Gaussian with the same first two moments; either way it equals the Gaussian chain's initial message, which has those same moments by construction.

Step. Suppose $\mu_{i-1}^{\rightarrow}$ agrees with the Gaussian chain's message, so it is Gaussian with some mean m and variance v. Multiplying by ℓ_{i-1} multiplies two Gaussian densities and yields a Gaussian, with mean and variance given by the usual precision-weighted combination — identical in both chains, since the likelihood is the same object. Taking the transition integral against $a_i = \rho a_{i-1} + \varepsilon_i$ convolves with φ and scales by ρ, which maps $(m, v) \mapsto (\rho m,\, \rho^2 v + q)$: the mean and variance of a convolution depend on φ *only* through its mean and variance, so this is the same map for every innovation with moments $(0, q)$, and in particular the same as for the Gaussian chain. In the Gaussian chain the convolution of two Gaussians is already Gaussian, so no projection occurs there; here Π_{G} is applied, and since the pre-projection function has exactly the moments just computed, the projection returns precisely the Gaussian chain's message.

The backward recursion runs the same argument with the transition map inverted, $(m, v) \mapsto (m/\rho,\, (v + q)/\rho^2)$, which is where $\rho \neq 0$ is used. Its base case needs one word of care. The terminal backward message is $r_n \equiv 1$, which is not a normalisable density and so is not in the domain of Π_{G}; the induction therefore starts one step earlier, at the terminal *unary likelihood* ℓ_n, which is Gaussian and normalisable. Every subsequent backward message is a likelihood-weighted transition integral of a normalisable Gaussian and stays in the domain. Nothing is lost by starting there, because $r_n \equiv 1$ contributes no factor: the first object the recursion actually projects is ℓ_n.

Hence every projected belief equals the corresponding Gaussian-chain belief, and the returned posterior mean is the Gaussian chain's posterior mean. For jointly Gaussian zero-mean $(\mathbf{a}, \mathbf{x})$ the conditional mean is linear in $\mathbf{x}$ and therefore coincides with the best linear estimator, which is (4): since $\mathbf{x} = e^{-t}\mathbf{a} + \sqrt{\Delta_t}\,\mathbf{z}$ with $\mathbf{z}$ independent, $\mathrm{Cov}(\mathbf{a}, \mathbf{x}) = e^{-t}\mathbf{\Sigma}_0$ and $\mathrm{Cov}(\mathbf{x}) = e^{-2t}\mathbf{\Sigma}_0 + \Delta_t \mathbf{I}$. Since the recursion never read φ beyond $(0, q)$, the same estimator is returned for every innovation law with those moments. $\square$